Domain and Range

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Domain from equations

State the domain of each function:

a $f(x) = \frac{1}{x-4}$

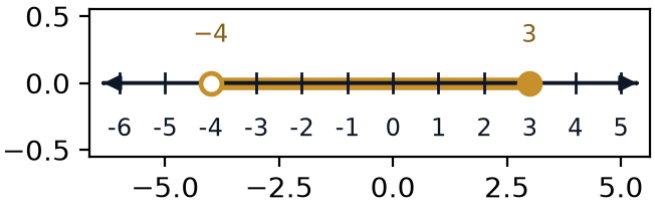
b $g(x) = \sqrt{2x-6}$

c $h(x) = \frac{x+1}{x^2-9}$

d $k(x) = \sqrt[3]{x}$

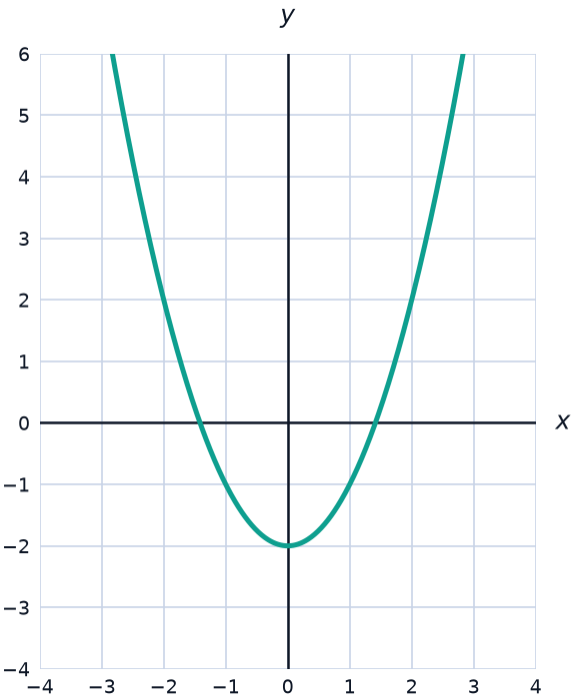
2 [Foundational] • Target: Domain from equations

Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.



3 [Foundational] • Target: Domain and range from graphs

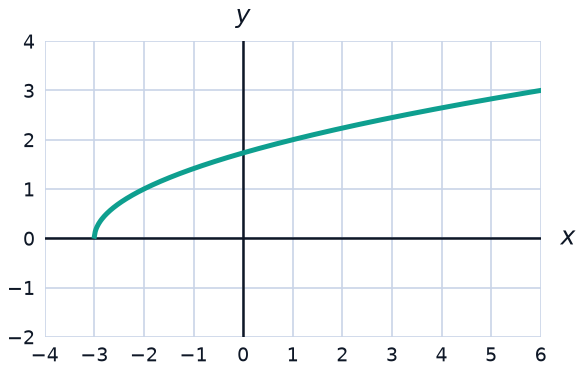
The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.



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4 [Foundational] • Target: Domain and range from graphs

The graph of $y = \sqrt{x + 3}$ is shown. State the domain and range of the function.

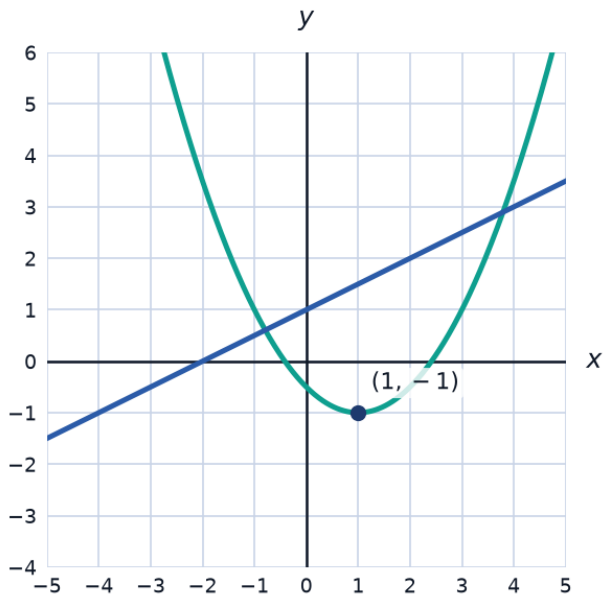


5 [Foundational] • Target: Finding the range algebraically

By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

6 [Foundational] • Target: Finding the range algebraically

Consider $f(x) = -2(x - 1)^2 + 5$.



a State the coordinates of the vertex.

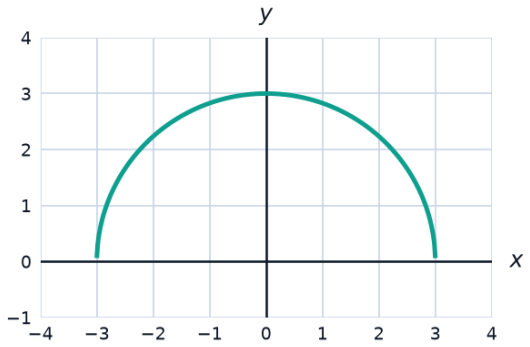
b Does the parabola open upward or downward?

c State the range of f.

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7 [Foundational] • Target: Finding the range algebraically

Find the range of $g(x) = \sqrt{9 - x^2}$ (the upper half of a circle).



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range in applied contexts

A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20 - x)$.

a State the domain of A in this context (both dimensions must be positive).

b Find $A(6)$ and $A(15)$.

c State the range of A over this domain, to the nearest whole number.

9 [Intermediate] • Target: Domain and range in applied contexts

An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12 - 2x)^2$.

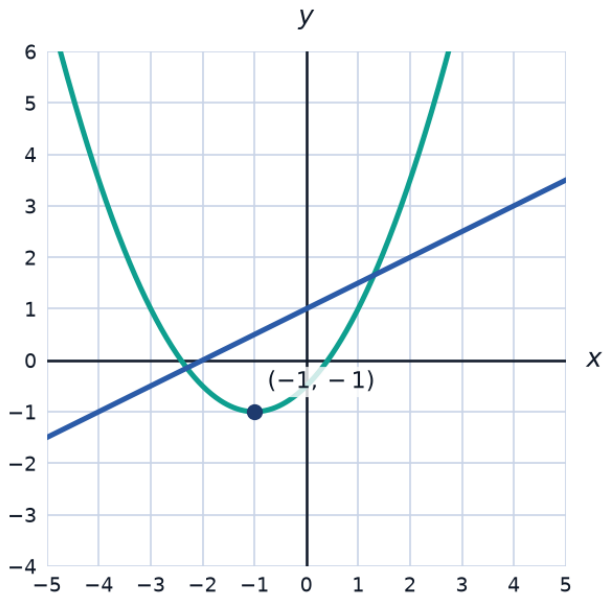
a State the domain of V in this physical context.

b Find $V(1)$ and $V(4)$.

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10 [Intermediate] • Target: Domain of composite and combined functions

Let $f(x) = \sqrt{x-2}$ and $g(x) = \frac{1}{x}$. Find the domain of:



a $(f+g)(x)$

b $(f \circ g)(x)$

11 [Intermediate] • Target: Domain of composite and combined functions

Find the domain of $f(x) = \frac{1}{\sqrt{x^2-4}}$, writing your answer in interval notation.

12 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $f(x) = \frac{x^2-4}{x^2-x-6}$, factoring the denominator first.

13 [Intermediate] • Target: Domain from rational functions with factoring

Find the domain of $g(x) = \frac{\sqrt{x+4}}{x-1}$, combining a root restriction with a denominator restriction.

14 [Intermediate] • Target: Range using transformations

The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

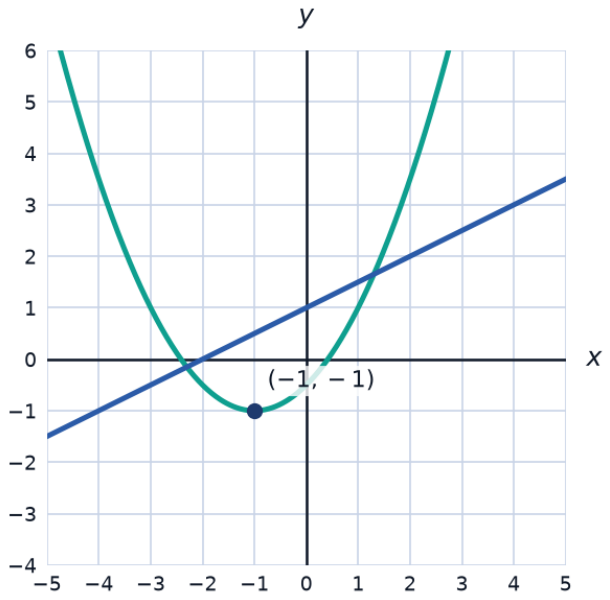
15 [Challenging] • Target: Range using transformations • Cross-Lesson: 5.03 Parametric Equations

The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.

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16 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.



17 [Challenging] • Target: Domain of composite and combined functions • Cross-Lesson: 6.02 Polar Equations and Graphs

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

18 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

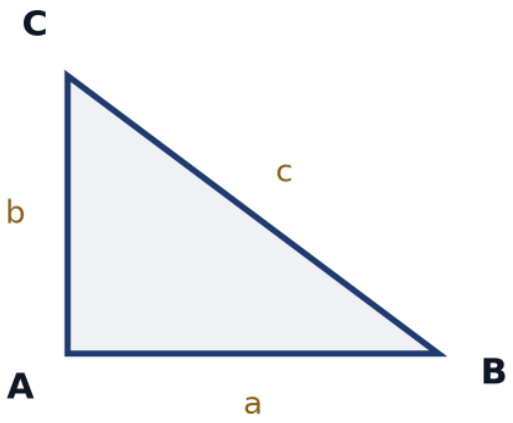
19 [Challenging] • Target: Range using transformations • Cross-Lesson: 7.02 Logarithms and Their Properties

What is the domain and range of $y = \tan^{-1}(x)$?

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20 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

In ΔABC, $a = 7$, $b = 9$, and $c = 10$. Find the measure of the smallest angle to the nearest tenth of a degree, and compute the area of ΔABC using Heron's formula.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $f(x) = \frac{x-3}{x^2-16}$.

a. $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$

b. $(-\infty, 4) \cup (4, \infty)$

c. $[-4, 4]$

d. All real numbers except $x = 3$

22 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $g(x) = \sqrt{2x-8}$.

a. $[4, \infty)$

b. $(4, \infty)$

c. $[8, \infty)$

d. $(-\infty, 4]$

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23 [Foundational] • Target: Finding Domains of Rational and Radical Functions

Find the domain of $h(x) = \frac{1}{\sqrt{9 - x^2}}$.

a. $(-3, 3)$

b. $[-3, 3]$

c. $(-\infty, -3) \cup (3, \infty)$

d. $[0, 3)$

24 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $f(x) = x^2 - 4x + 7$.

a. $[3, \infty)$

b. $[7, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

25 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $g(x) = -|x + 2| + 5$.

a. $(-\infty, 5]$

b. $[5, \infty)$

c. $(-\infty, -2]$

d. $[-5, 5]$

26 [Foundational] • Target: Finding Ranges of Algebraic Functions

Find the range of $h(x) = \sqrt{25 - x^2}$.

a. $[0, 5]$

b. $[-5, 5]$

c. $[0, 25]$

d. $[5, \infty)$

27 [Foundational] • Target: Piecewise Functions

Evaluate $f(-2)$ for $f(x)$ defined as $2x + 1$ for $x < 0$, and $x^2 - 3$ for $x \geq 0$.

a. -3

b. 1

c. -2

d. 7

28 [Foundational] • Target: Piecewise Functions

Find the domain of $f(x)$ defined as $\frac{1}{x + 3}$ for $x < 1$, and $\sqrt{x - 1}$ for $x \geq 1$.

a. $(-\infty, -3) \cup (-3, \infty)$

b. $[1, \infty)$

c. All real numbers

d. $(-\infty, 1) \cup [1, \infty)$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Piecewise Functions**

Determine the range of $g(x)$ defined as $-x$ for $x < 0$, and $x + 1$ for $x \geq 0$.

a. $(0, \infty)$

b. $[1, \infty)$

c. $[0, \infty)$

d. All real numbers

30 **[Intermediate]** • **Target: Domain and Range of Compositions**

If $f(x) = \frac{1}{x-2}$ and $g(x) = \sqrt{x}$, find the domain of $(f \circ g)(x)$.

a. $[0, 4) \cup (4, \infty)$

b. $[0, \infty)$

c. $(4, \infty)$

d. All reals except $x = 2$

31 **[Intermediate]** • **Target: Domain and Range of Compositions**

If $f(x) = x^2$ and $g(x) = 3x - 1$, what is the range of $(f \circ g)(x)$ for $x \in \mathbb{R}$?

a. $[0, \infty)$

b. $[-1, \infty)$

c. $[1, \infty)$

d. All real numbers

32 **[Intermediate]** • **Target: Domain and Range of Compositions**

Find the domain of $(g \circ f)(x)$ if $f(x) = 2x + 6$ and $g(x) = \ln(x)$.

a. $(-3, \infty)$

b. $[0, \infty)$

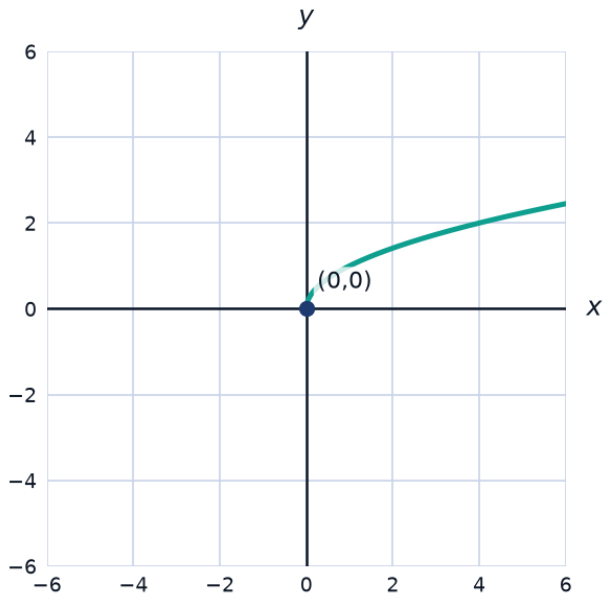
c. $(-6, \infty)$

d. $(3, \infty)$

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33 [Intermediate] • Target: Domain from equations

If $f(x) = \sqrt{x - 0}$ and $g(x) = x^2 - 1$, what is the domain of $f(g(x))$?



a. All real numbers

b. $x \leq -1$ or $x \geq 1$

c. $x \geq 1$

d. $-1 \leq x \leq 1$

34 [Intermediate] • Target: Domain and range from graphs

What is the domain and range of $y = \tan^{-1}(x)$? (variant 9)

a. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

b. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

c. Domain $(-\infty, \infty)$, range $[0, \pi]$

d. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

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35 **[Intermediate]** • **Target: Finding the range algebraically**

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

36 **[Intermediate]** • **Target: Domain and range in applied contexts**

What is the domain and range of $y = \sin^{-1}(x)$? (variant 2)

a. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

b. Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

c. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

d. Domain $[-1, 1]$, range $(-\infty, \infty)$

37 **[Intermediate]** • **Target: Domain of composite and combined functions**

What is the domain and range of $y = \sin^{-1}(x)$? (variant 10)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

d. Domain $(-\infty, \infty)$, range $[-\frac{\pi}{2}, \frac{\pi}{2}]$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Domain from rational functions with factoring • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

What is the domain and range of $y = \sin^{-1}(x)$? (variant 6)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

c. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

39 [Challenging] • Target: Range using transformations • Cross-Lesson: 7.02 Logarithms and Their Properties

What is the domain and range of $y = \sin^{-1}(x)$? (variant 8)

a. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

b. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

c. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

d. Domain $[-1, 1]$, range $(-\infty, \infty)$

40 [Challenging] • Target: Domain and range of piecewise functions • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

What is the domain and range of $y = \sin^{-1}(x)$? (variant 7)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

c. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

d. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

41 [Challenging] • Target: Finding Domains of Rational and Radical Functions • Cross-Lesson: 7.04 Applications: Growth and Decay

What is the domain of $f(x) = \sqrt{x - 4}$?

a. $x > 4$

b. $x \geq 4$

c. $x \geq -4$

d. $x \leq 4$

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42 [Challenging] • Target: Finding Ranges of Algebraic Functions • Cross-Lesson: 8.01 Rational Functions and Asymptotes

What is the domain and range of $y = \cos^{-1}(x)$? (variant 5)

a. Domain $(-\infty, \infty)$, range $[0, \pi]$

b. Domain $[-1, 1]$, range $[0, \pi]$

c. Domain $[-1, 1]$, range $(-\infty, \infty)$

d. Domain $[0, \pi]$, range $[-1, 1]$

43 [Challenging] • Target: Piecewise Functions • Cross-Lesson: 8.02 Graphing Rational Functions

Using the Remainder Theorem, find the remainder when $f(x) = 2x^2 - 1x + 6$ is divided by $x + 3$.

a. -27

b. 28

c. 27

d. 26

44 [Challenging] • Target: Domain and Range of Compositions • Cross-Lesson: 8.03 Introduction to Limits

What is the range of $f(x) = (x - 4)^2 + 3$?

a. $y \leq 3$

b. $y \geq 4$

c. All real numbers

d. $y \geq 3$

45 [Challenging] • Target: Domain from equations • Cross-Lesson: 9.01 Arithmetic Sequences and Series

What is the domain of $f(x) = \frac{x + 1}{(x + 3)(x + 1)}$?

a. All real numbers

b. All reals except $x = -3$

c. All reals except $x = -3$ and $x = -1$

d. All reals except $x = -4$

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